```
> restart;#Audion, the phonon Planck's law;
> ec := 1.602176634*10^(-19);
  va := 343;
  c := 299792458;
```

$$ec := 1.602176634 \times 10^{-19}$$

$$va := 343$$

$$c := 299792458 \quad (1)$$

```
> E1 := 0.00762014687*t + a*t^2 + b*t^3 + d*t^4 + f*t^5;
```

$$E1 := 0.00762014687\,t + a\,t^2 + b\,t^3 + d\,t^4 + f\,t^5 \quad (2)$$

```
> E2 := subs(t = 435, E1);
```

$$E2 := 3.314763888 + 189225\,a + 82312875\,b + 35806100625\,d + 15575653771875\,f \quad (3)$$

```
> E3 := subs(t = 479, E1);
```

$$E3 := 3.650050351 + 229441\,a + 109902239\,b + 52643172481\,d + 25216079618399\,f \quad (4)$$

```
> E4 := subs(t = 528, E1);
```

$$E4 := 4.023437547 + 278784\,a + 147197952\,b + 77720518656\,d + 41036433850368\,f \quad (5)$$

```
> E5 := subs(t = 2860, E1);
```

$$E5 := 21.79362005 + 8179600\,a + 23393656000\,b + 66905856160000\,d + 191350748617600000\,f \quad (6)$$

```
> a := -1.44670357887361*10^(-7);b := -1.049267156*10^(-9); d :=
  1.890440485*10^(-12);f :=-6.233924848*10^(-16);
```

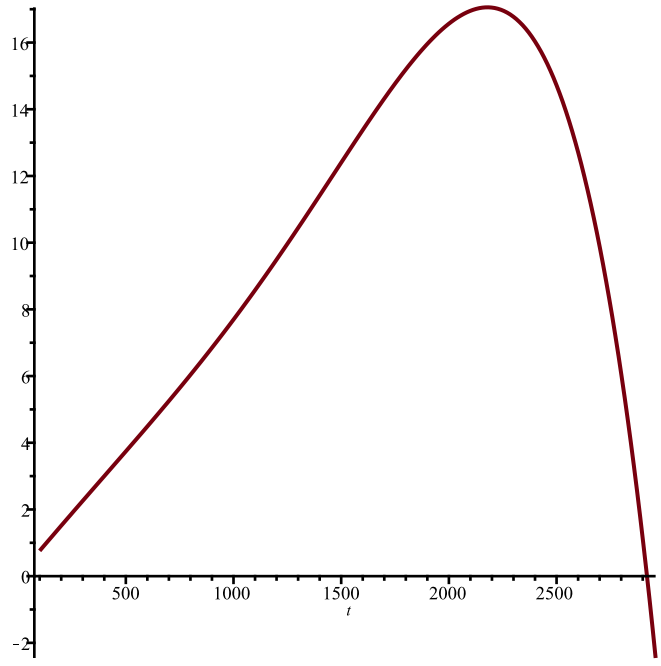
$$a := -1.446703579 \times 10^{-7}$$

$$b := -1.049267156 \times 10^{-9}$$

$$d := 1.890440485 \times 10^{-12}$$

$$f := -6.233924848 \times 10^{-16} \quad (7)$$

```
> plot(E1,t=100..2960);solve(3.259 = E1, t);solve(3.59 = E1, t);
  solve(3.95 = E1, t);E1;
```



434.9999999, 2860.000000, 579.9964863 + 1617.918467 I, −1422.488521, 579.9964863

$$-1617.918467\,I$$

$$479.6275541,\ 2853.405326,\ 565.9474392 + 1617.810863\,I,\ -1432.423307,\ 565.9474392 - 1617.810863\,I$$

$$528.0000000,\ 2846.108503,\ 550.6640041 + 1618.311176\,I,\ -1442.932060,\ 550.6640041 - 1618.311176\,I$$

$$0.00762014687\,t - 1.446703579\times10^{-7}\,t^2 - 1.049267156\times10^{-9}\,t^3 + 1.890440485\times10^{-12}\,t^4 - 6.233924848\times10^{-16}\,t^5 \quad (8)$$

```
> restart;#Phonon
> with(PDEtools, dchange):
> Digits:=30;
```

$$Digits := 30 \quad (9)$$

```
> N:=8*Pi*nu^2*E(nu)/(c^3*(exp(E(nu)*c^2/(v^2*k*T)) - 1));V:=4/3*
  Pi*rb^3;
```

$$N := \frac{8\,\pi\,\nu^2\,E(\nu)}{c^3\left(e^{\frac{E(\nu)\,c^2}{\nu^2\,kT}} - 1\right)}$$

$$V := \frac{4\,\pi\,rb^3}{3} \quad (10)$$

```
> ET:=dchange({nu=1/t},N*V,expand);#Change of var in t = period of
  a single oscilation;
```

$$ET := \frac{32\,\pi^2\,E(t)\,rb^3}{3\,t^2\,c^3\left(e^{\frac{E(t)\,c^2}{\nu^2\,kT}} - 1\right)} \quad (11)$$

```
> SolT:=subs(t=t,(ET)-subs(t=t+Tq,(ET)));#We do the calculus per
  one period Tq;
```

$$SolT := \frac{32\,\pi^2\,E(t)\,rb^3}{3\,t^2\,c^3\left(e^{\frac{E(t)\,c^2}{\nu^2\,kT}} - 1\right)} - \frac{32\,\pi^2\,E(t+Tq)\,rb^3}{3\,(t+Tq)^2\,c^3\left(e^{\frac{E(t+Tq)\,c^2}{\nu^2\,kT}} - 1\right)} \quad (12)$$

```
> Sol1:= dchange({t=1/nu},SolT,params=Tq,expand);#We change back to
  the solution in frequency nu with approx E(nu);
```

$$Sol1 := \frac{32\,\pi^2\,\nu^2\,E(\nu)\,rb^3}{3\,c^3\left(e^{\frac{E(\nu)\,c^2}{\nu^2\,kT}} - 1\right)} - \frac{32\,\pi^2\,E(\nu)\,rb^3}{3\left(\frac{1}{\nu} + Tq\right)^2 c^3\left(e^{\frac{E(\nu)\,c^2}{\nu^2\,kT}} - 1\right)} \quad (13)$$

```
> solve(E(nu)*A=Sol1,E(nu));#  the energy of one photon in
  approximation E(nu+1/Tq):
```

$$\frac{\ln\left(\frac{32\,\pi^2\,Tq^2\,\nu^4\,rb^3 + 64\,\pi^2\,Tq\,\nu^3\,rb^3 + 3\,A\,Tq^2\,c^3\,\nu^2 + 6\,A\,Tq\,c^3\,\nu + 3\,A\,c^3}{3\,A\,c^3\left(Tq^2\,\nu^2 + 2\,Tq\,\nu + 1\right)}\right)\nu^2\,k\,T}{c^2} \quad (14)$$

```
> E(nu):=%;
```

$$E(\nu) := \frac{\ln\left(\frac{32\,\pi^2\,Tq^2\,\nu^4\,rb^3 + 64\,\pi^2\,Tq\,\nu^3\,rb^3 + 3\,A\,Tq^2\,c^3\,\nu^2 + 6\,A\,Tq\,c^3\,\nu + 3\,A\,c^3}{3\,A\,c^3\left(Tq^2\,\nu^2 + 2\,Tq\,\nu + 1\right)}\right) v^2\,k\,T}{c^2} \tag{15}$$

```
> series((1/nu+Tq)^(-1),nu);# Approximation
```

$$\nu - Tq\,\nu^2 + Tq^2\,\nu^3 - Tq^3\,\nu^4 + Tq^4\,\nu^5 + O\left(\nu^6\right) \tag{16}$$

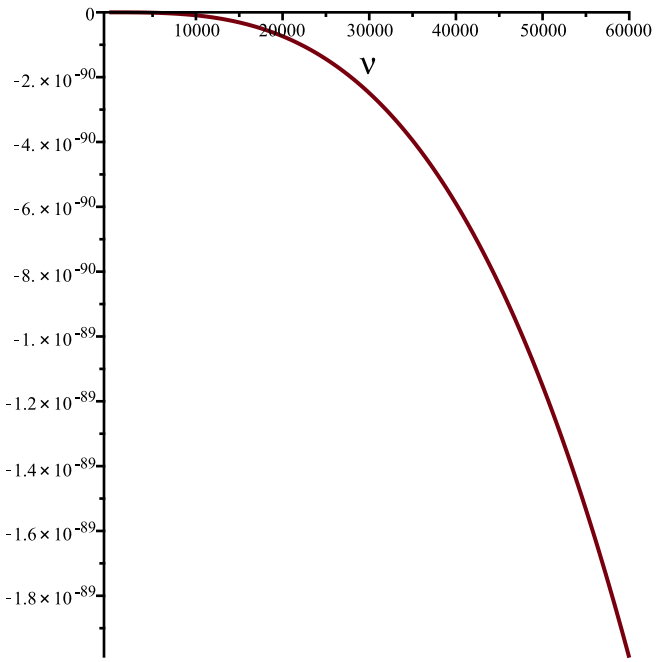
```
> E0:=series(E(nu),nu,9);
```

$$E0 := \frac{64\,\pi^2\,Tq\,rb^3\,v^2\,k\,T}{3\,A\,c^5}\,\nu^3 - 32\,\frac{\pi^2\,Tq^2\,rb^3\,v^2\,k\,T}{A\,c^5}\,\nu^4 + \frac{128}{3}\,\frac{\pi^2\,Tq^3\,rb^3\,v^2\,k\,T}{A\,c^5}\,\nu^5$$
$$+ \frac{\left(-\frac{160\,\pi^2\,Tq^4\,rb^3}{3\,A\,c^3} - \frac{2048\,\pi^4\,Tq^2\,rb^6}{9\,A^2\,c^6}\right) v^2\,k\,T}{c^2}\,\nu^6$$
$$+ \frac{\left(\frac{64\,\pi^2\,Tq^5\,rb^3}{A\,c^3} + \frac{2048\,\pi^4\,Tq^3\,rb^6}{3\,A^2\,c^6}\right) v^2\,k\,T}{c^2}\,\nu^7$$
$$+ \frac{\left(-\frac{224\,\pi^2\,Tq^6\,rb^3}{3\,A\,c^3} - \frac{12800\,\pi^4\,Tq^4\,rb^6}{9\,A^2\,c^6}\right) v^2\,k\,T}{c^2}\,\nu^8 + O\left(\nu^9\right) \tag{17}$$

```
> ec:=1.602176634*10^(-19);k:=1.3806490000*10^(-23);rb:=5.239*10^
  (-11);v:=343;h := 6.626070150*10^(-34);Tq:=rb/v;c:=299792458;T:=
  273.15+24;A:=-10^4;#h1:=3.960517553*10^(-34)*ec;h2:=-6.133245132*
  10^(-49)*ec;
```

$$ec := 1.60217663400000000000000000000 \times 10^{-19}$$
$$k := 1.38064900000000000000000000000 \times 10^{-23}$$
$$rb := 5.23900000000000000000000000000 \times 10^{-11}$$
$$v := 343$$
$$h := 6.62607015000000000000000000000 \times 10^{-34}$$
$$Tq := 1.52740524781341107871720116618 \times 10^{-13}$$
$$c := 299792458$$
$$T := 297.15$$
$$A := -10000 \tag{18}$$

```
> plot(E0,nu=10..60000);
```



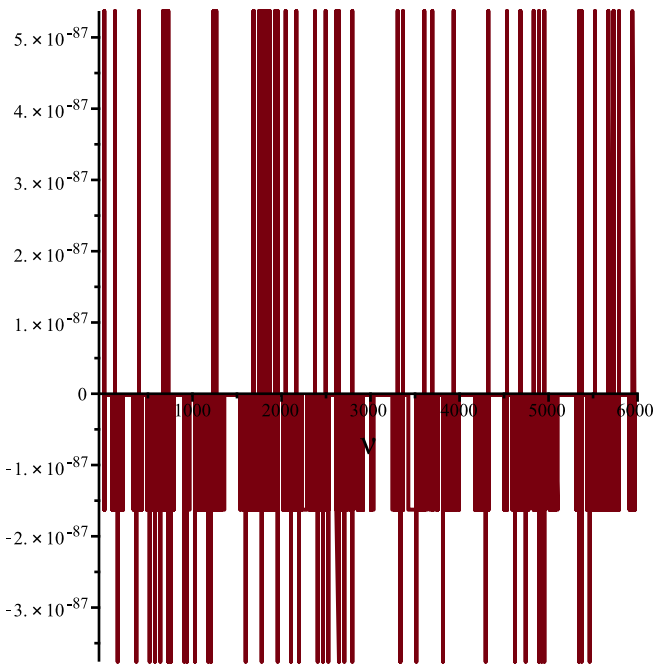

```
> ALT := simplify(eval(E(nu)));
```

$$ALT := 5.37039032160737846380942952821 \times 10^{-33} \ln\left(\left(-1.31075074237509638052271274251 \times 10^{-84} \nu^4 - 1.71631038226630295292723981937 \times 10^{-71} \nu^3 + 2.33296679104794770886280376375 \times 10^{-26} \nu^2 + 3.05481049562682215743440233236 \times 10^{-13} \nu + 1.\right) \Big/ \left(2.33296679104794770886280376374 \times 10^{-26} \nu^2 + 3.05481049562682215743440233236 \times 10^{-13} \nu + 1.\right)\right) \quad (19)$$

```
> Digits:=55;
```

$$Digits := 55 \quad (20)$$

```
> plot(E(nu),nu=10..6000);
```




```
> plot(ALT,nu=10..60000, thickness=3);
```

